Assignment 2:
Data Analysis Report

36103: Statistical Thinking for Data Science

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Executive Summary

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Introduction

1 Introduction

1.1 Problem Statement

Electricity supply and demand must balance every 30 minutes. When they don't, grid operators absorb the cost. NSW wholesale prices can reach $17,500/MWh under the current market cap (Australian Energy Regulator [AER], 2025). Accurate short-term demand forecasting is what keeps dispatch decisions ahead of those spikes.

1.2 Rationale

The National Electricity Market (NEM) serves around 11 million customers across eastern Australia, with NSW consuming approximately 40% of total NEM generation (AER, 2025; National Electricity Market, 2025). Demand is not static; it shifts every 30 minutes with temperature, time of day, and day type, and the grid has no meaningful storage to absorb mismatches.

The supply side is also changing. Australia now ranks first globally in per capita solar electricity generation (Clean Energy Council [CEC], 2025), with NSW holding the highest total installed rooftop capacity of any state at 6.6 GW (PV Magazine, 2024). Rooftop solar suppresses midday grid demand but drops off sharply in late afternoon, creating a steep evening ramp that dispatchable generators must cover at short notice. NSW already sees a 34-41% gap between its daily minimum and maximum demand, among the largest swings in the NEM (AER, 2025).

Prices reflect that volatility. In 2024, NSW average wholesale electricity prices rose 43% year-on-year to $150/MWh (AER, 2025). Forecasting errors at that price level carry direct costs for operators, retailers, and consumers.

1.3 Project Aims and Objectives

Methodology

2 Methodology

2.1 Modelling

2.1.1 Linear Regression (Ridge)

2.1.2 Gradient Boosting Trees

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